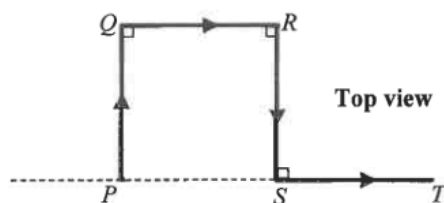


1. A well-insulated vessel with negligible heat capacity contains warm water at 45°C . The temperature of warm water drops by 5°C after mixing with 50 g water at temperature 0°C added to the vessel. Find the original mass of warm water in the vessel.
- A. 400 g
 B. 450 g
 C. 500 g
 D. 550 g

2. When ice melts into water at 0°C , what happens to the molecules during the melting process?
- (1) their average separation increases
 (2) their average kinetic energy increases
 (3) their average potential energy increases
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

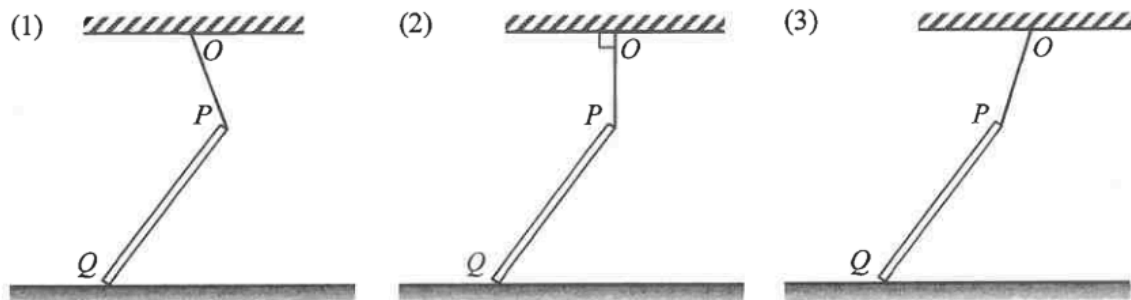
- *3. A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27°C . Find the gas pressure in the balloon. Given: mass of one mole of helium gas = $4 \times 10^{-3}\text{ kg}$
- A. $9.93 \times 10^4\text{ Pa}$
 B. $1.00 \times 10^5\text{ Pa}$
 C. $1.05 \times 10^5\text{ Pa}$
 D. $1.09 \times 10^5\text{ Pa}$

4. A car travels with constant speed v along a horizontal road $PQRST$ with four sections of equal length as shown. Determine v if the car's average velocity for the journey $PQRST$ is 20 km h^{-1} in magnitude.



- A. 10 km h^{-1}
 B. 20 km h^{-1}
 C. 40 km h^{-1}
 D. It cannot be found as the length of each section is unknown.

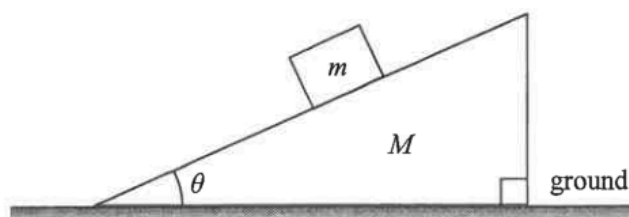
5. In each figure below, a uniform rod PQ has its upper end P attached to point O on the ceiling by a light inextensible string. The rod's lower end Q rests on a rough horizontal ground as shown.



In which figure is the frictional force acting on the rod due to the ground towards the right?

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

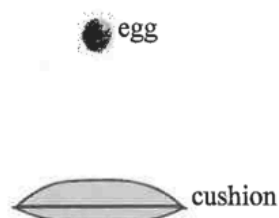
6.



A block of mass m is placed on a wedge of mass M as shown. The system remains at rest. What is the normal force exerted on the wedge by the ground?

- A. Mg
- B. $(M + m)g$
- C. $Mg + mg \cos \theta$
- D. $Mg + mg \tan \theta$

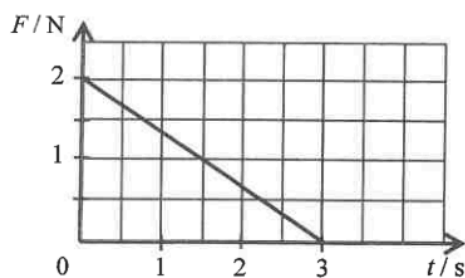
7. An egg is released from a height and falls towards a cushion without breaking.



Which of the following is the most probable explanation?

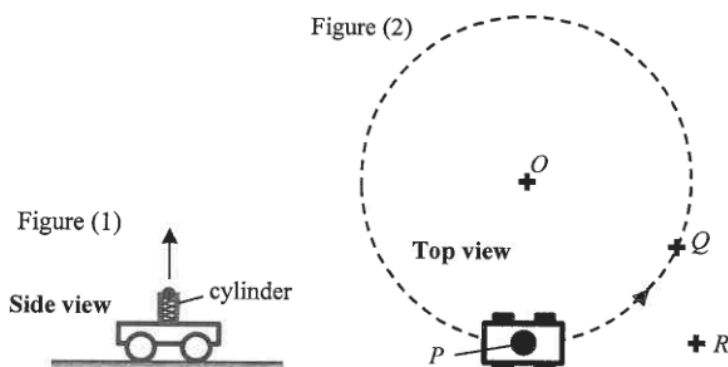
- A. The cushion makes the egg stop at a shorter distance.
- B. The cushion helps lengthen the time of impact.
- C. The cushion helps reduce part of the gravitational force acting on the egg.
- D. The cushion increases the reaction force acting on the egg during impact.

8. An object of mass 2 kg moves with velocity 1 m s^{-1} initially. It is acted upon by a force F which varies with time t as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at $t = 3\text{ s}$.



- A. 1.5 m s^{-1}
- B. 2.5 m s^{-1}
- C. 3.5 m s^{-1}
- D. 4.5 m s^{-1}

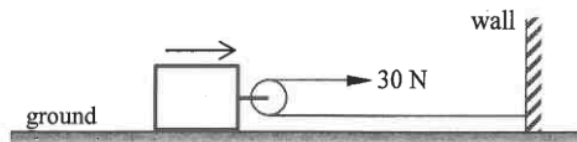
- *9. A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).



The ball is projected when the trolley is at point P . After some time, the ball falls back to the ground as the trolley just reaches point Q . Neglecting air resistance, where would the ball land?

- A. O
- B. P
- C. Q
- D. R

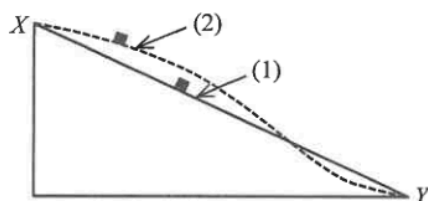
10.



On a horizontal ground there is a block attached with a smooth light pulley as shown. A light and inextensible horizontal string, with one end fixed to a wall, passes the pulley. A man exerts a horizontal force of 30 N at the other end of the string and pulls a distance of 4 m . Find the work done by the man if the ground exerts a frictional force of 10 N on the block.

- A. 20 J
- B. 80 J
- C. 100 J
- D. 120 J

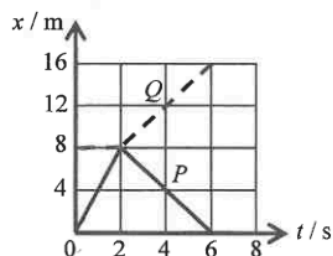
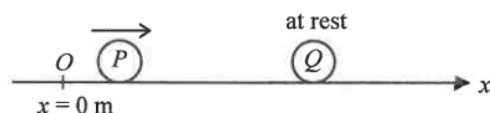
11. In the figure below, the straight path (1) and the curved path (2) in a vertical plane are smooth. A small block slides from rest at X along the respective paths to Y .



Which of the following about the speed of the block at Y and the time taken to reach Y is correct? Neglect air resistance.

	speed at Y	time taken to reach Y
A.	the same	different
B.	the same	the same
C.	different	the same
D.	different	different

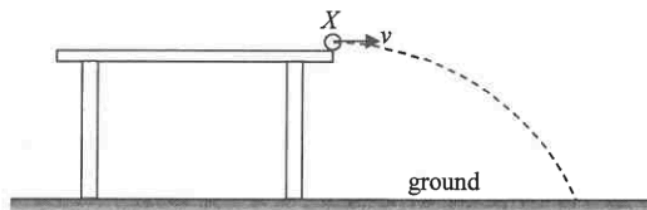
12.



On a smooth horizontal surface, sphere P travelling along the x -axis collides head-on with another sphere Q , which is initially at rest at $x = 8$ m. The graph shows the displacement-time (x - t) relationship of P (solid line) and Q (dotted line). The collision takes place at $t = 2$ s and the collision time is negligible. Which statement is correct?

- A. The collision is perfectly inelastic.
- B. The mass of P is larger than that of Q .
- C. P still moves along $+x$ direction after collision.
- D. P and Q move with the same speed after collision.

- *13. A marble leaves the edge of a table at point X with a horizontal speed v and hits the ground at a certain point as shown.

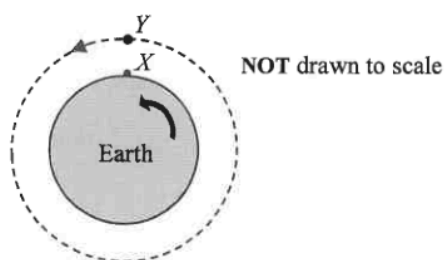


If the marble leaves the table at a higher speed, which of the following would remain unchanged? Neglect air resistance.

- (1) the time of flight of the marble in the air
- (2) the acceleration of the marble during its flight in air
- (3) the horizontal distance between X and the landing point

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

- *14. In the figure, X is an object resting on the Earth's equator. Y is a geostationary satellite moving in a circular orbit above the equator such that it always appears stationary to an observer on Earth.

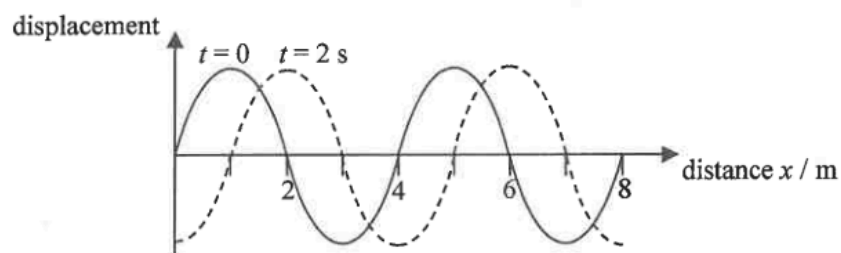


Which descriptions about the motion of X and Y are correct?

- (1) The motion of X and Y takes the same period.
- (2) X is moving with a slower speed.
- (3) X has a greater acceleration.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

15. The figure shows the displacement-distance graph at time $t = 0$ and $t = 2$ s respectively of a wave travelling to the right.

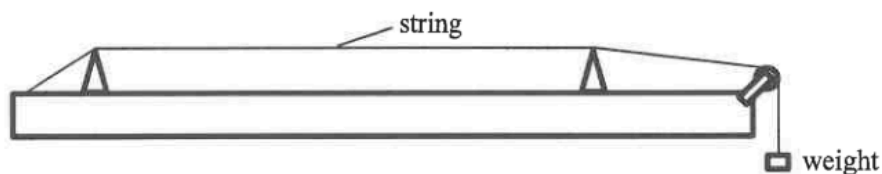


Which statements about the wave are correct ?

- (1) The wavelength is 4 m.
- (2) The period is 4 s.
- (3) The speed of the wave can be 2.5 m s^{-1} .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

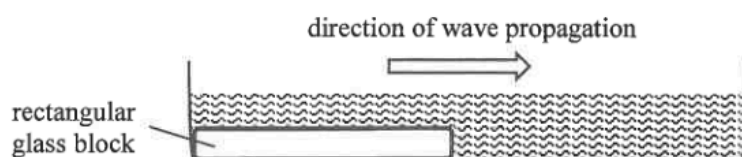
16. Referring to the set-up below, which of the following combinations would give the highest speed of wave propagation along the string when the string is plucked ?



tension of the string radius of the cross-section of the string

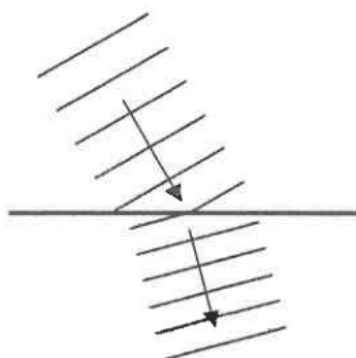
- | | | |
|----|------|------|
| A. | T | r |
| B. | T | $2r$ |
| C. | $2T$ | r |
| D. | $2T$ | $2r$ |

17. Plane water waves travel from shallow water to deep water in a ripple tank as shown below.

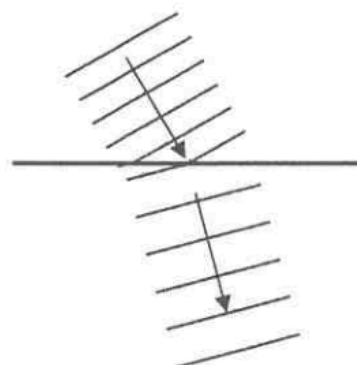


Which diagram correctly shows the top view of the wavefronts in the ripple tank ?

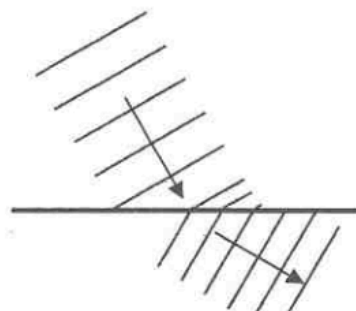
A.



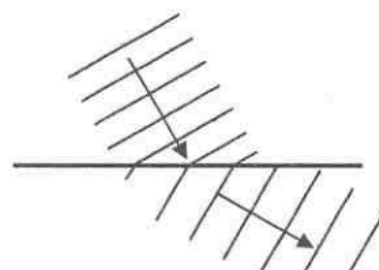
B.



C.

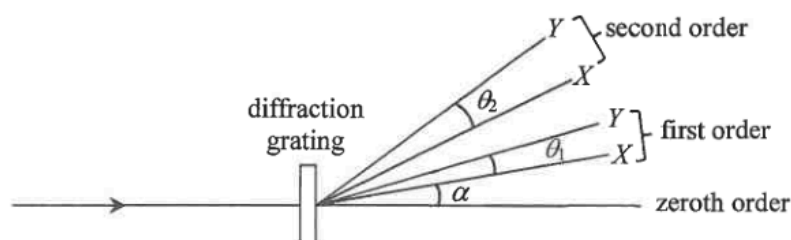


D.



2022 Q18

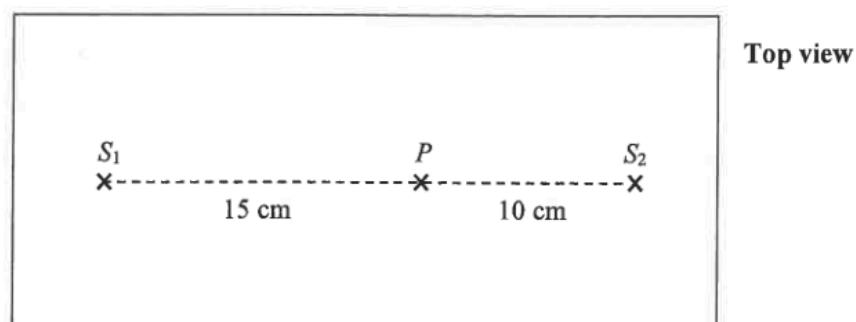
*18.



A light beam which consists of monochromatic lights X and Y is incident normally on a diffraction grating. The figure shows the spectra of the first two orders. The diffraction angle of X in the first order is α . The angular separation between X and Y in the first order is θ_1 and that in the second order is θ_2 . Which statement **must be** correct ?

- A. The wavelength of Y is shorter than that of X .
- B. α would be larger if the grating spacing is smaller.
- C. θ_1 is independent of the grating spacing.
- D. $\theta_2 = 2\theta_1$

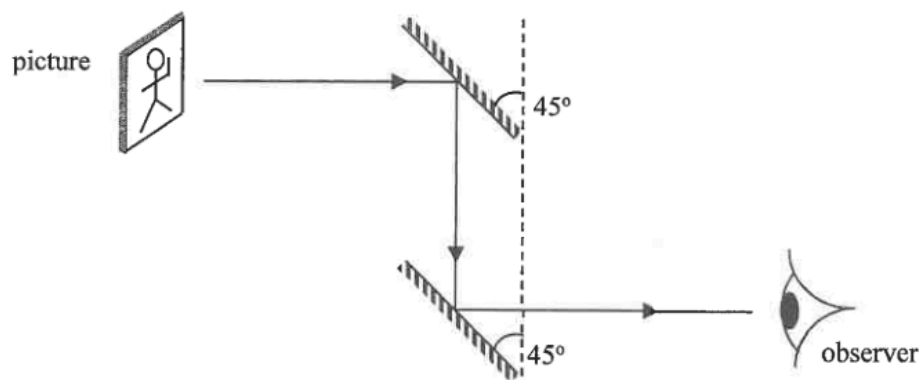
19. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase with frequency f and generate two sets of waves towards each other at a speed of 20 cm s^{-1} . Destructive interference occurs at point P on the straight line S_1S_2 .



Which of the following is a possible value of f ?

- A. 24 Hz
- B. 20 Hz
- C. 18 Hz
- D. 16 Hz

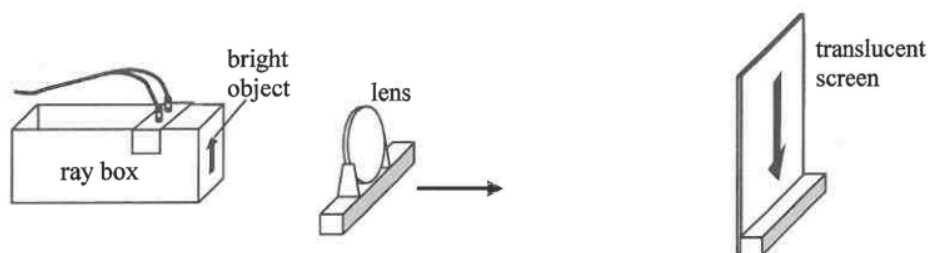
20. A periscope is formed by two plane mirrors as shown.



Which image will the observer see?

- A.
- B.
- C.
- D.

21. In the set-up below, the separation between the bright object and the translucent screen is fixed. A lens is moved from the object towards the screen as shown.



The first sharp image formed is inverted as shown and its length is 9 cm. When the lens is moved further towards the screen, a second sharp image of length 1 cm is observed. Which of the following statements is/are correct?

- (1) The second image is erect.
- (2) The length of the object is 3 cm.
- (3) There are at most two positions of the lens that can give a sharp image on the screen when the lens is moved.

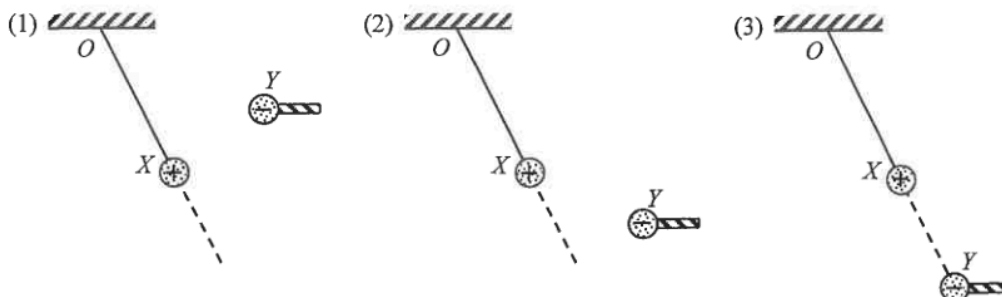
- A. (1) only
- B. (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

22. Which statements about ultrasound are correct?

- (1) Ultrasound is longitudinal waves.
- (2) Ultrasound needs media to propagate.
- (3) The speed of ultrasound in glass is higher than that in air.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

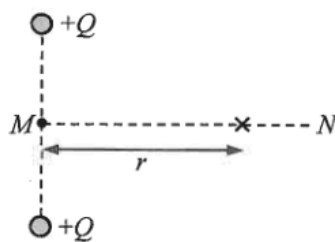
23. A positively charged sphere X of mass m is suspended from a fixed point O by a nylon thread. Another negatively charged sphere Y at the end of an insulating rod is held in various positions as shown. O , X and Y are in the same vertical plane.



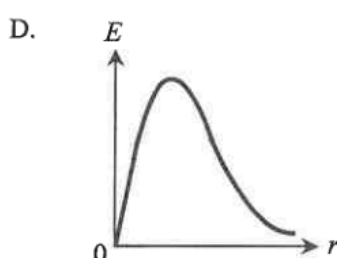
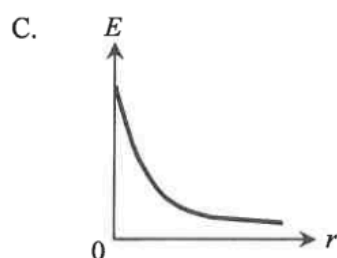
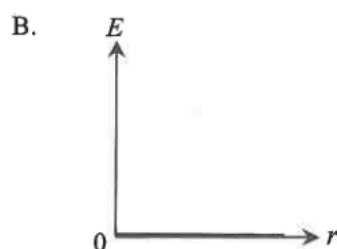
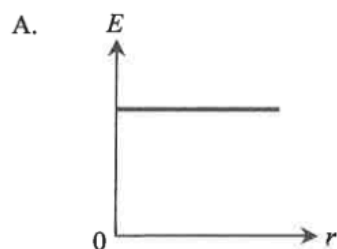
In which situation(s) could X be kept at rest as shown?

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

*24.



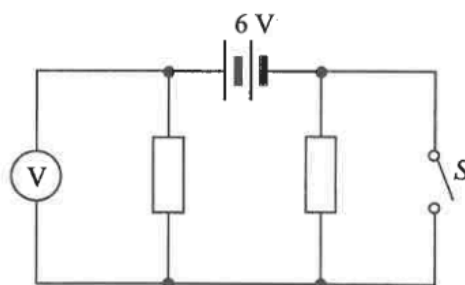
Two positive point charges $+Q$ are fixed as shown above. MN is the perpendicular bisector of the line joining the charges. Which graph correctly shows the variation of the electric field strength E on line MN with distance r from M ?



25. An electrical device is charged with a steady d.c. of 60 mA for 30 minutes. Find the **number of electrons** that pass through the device during charging.

- A. 108
 B. 1800
 C. 3.75×10^{17}
 D. 6.75×10^{20}

26.

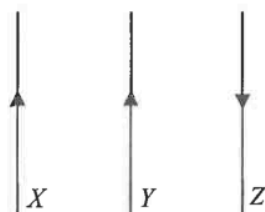


In the above circuit, the resistors are identical and the 6 V battery has negligible internal resistance. Which of the following is the set of readings on the voltmeter when (1) S is open; (2) S is closed ?

	S open	S closed
A.	0 V	6 V
B.	3 V	6 V
C.	6 V	0 V
D.	6 V	3 V

2022 Q27

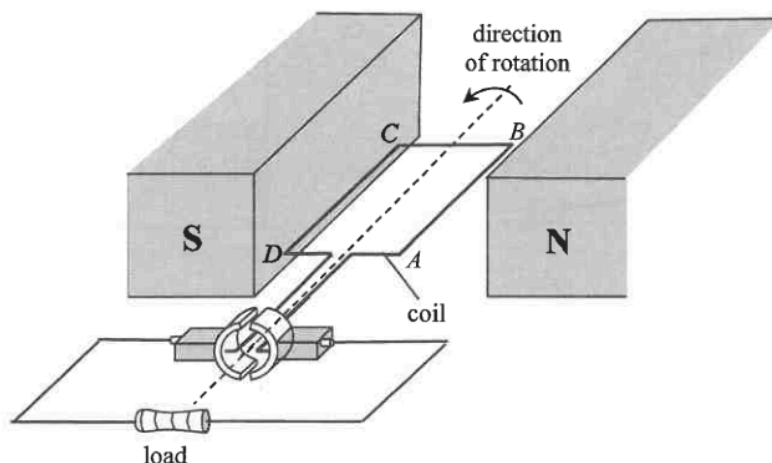
27. The figure below shows three parallel wires carrying currents in the directions shown.



If one of the wires experiences zero resultant magnetic force, that wire

- A. must be X .
- B. must be Y .
- C. must be Z .
- D. may be Y or Z .

28.



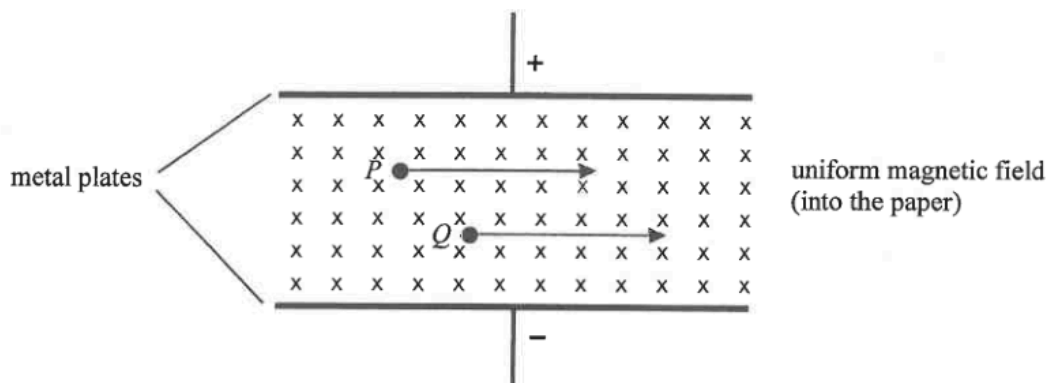
The figure shows the structure of a simple generator. Which of the following statements is/are correct ?

- (1) The magnetic force acting on side AB of the coil is upward at the instant shown.
- (2) The commutator would reverse the direction of current in the coil whenever the coil passes its vertical position.
- (3) The current flowing through the load is an unsteady d.c.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1) and (3) only

2022 Q29

- *29. Charged particles P and Q are moving in a region with mutually perpendicular uniform electric and magnetic fields as shown.



If both particles are not deflected by the fields, which of the following statements **must be** correct ? Neglect the effects of gravity.

- (1) They are positively charged.
- (2) They are moving with the same velocity.
- (3) They have the same charge to mass ratio.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

- *30. Arrange the following in ascending order of power delivered when each of them is connected to the same resistor.

- (1) a 100 Hz sinusoidal a.c. of peak voltage 2 V
- (2) a 50 Hz sinusoidal a.c. of r.m.s. voltage 2 V
- (3) a steady d.c. of voltage 1.5 V

- A. (1) (3) (2)
- B. (2) (3) (1)
- C. (1) (2) (3)
- D. (2) (1) (3)

2022 Q31

31. Which of the following statements about X-rays is **INCORRECT** ?

- A. X-rays can be produced by high-speed electrons hitting a metal target.
- B. X-rays are a kind of electromagnetic waves.
- C. X-rays can be deflected by an electric field.
- D. Although X-rays do not carry charges, they can cause ionization.

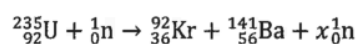
2022 Q32

- *32. Radioactive sources X and Y have initial activities 100 kBq and 200 kBq respectively. One day later, their activities are found to be equal. After another day, the activity of X becomes 80 kBq. Deduce the corresponding activity of Y .

- A. 40 kBq
- B. 50 kBq
- C. 89 kBq
- D. 160 kBq

2022 Q33

33. What is the value of x for the fission reaction below ?



- A. 1
- B. 2
- C. 3
- D. 4